Cell type distribution and axon projection mapping - data acquisition and processing

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Abstract

Even though scientists have been captivated by the diversity of brain cells for well over a century, since the initial anatomical descriptions by Santiago Ramon y Cajal, it is only recently that the development of new technologies allowed us to truly appreciate the astonishing complexity of cell types in the mammalian brain. Yet, even today, our knowledge of cell type anatomy and function is largely limited to a few brain areas, such as sensory cortex or hippocampus. This lack of whole-brain data has led to generalizations from one brain area to other regions with similar architecture. However, our preliminary studies applying quantitative whole-brain analysis have revealed unexpected region-specific differences in cell type ratios, and cell-type-specific axon projection patterns. These cell-type distribution and axon projection mapping can inform our understanding of brain development, wiring and function. We thus believe that closing the gaps in our knowledge of the cellular architecture of the brain will be a major contribution to enable a modern quantitative understanding of the brain as a system. Here we propose to establish the “Anatomy Segment”, a part of “A Comprehensive Center for Mouse Brain Cell Atlas”, and will specifically focus on Systematic Atlasing of Cell-Type Distribution and Axon Projections. This Segment will apply standardized, unbiased and largely automated methods developed in our laboratories to atlas the distribution and axon projection patterns of >35 molecularly defined projection neuron subpopulations and cell types that underlie major forebrain pathways. This work will yield the most comprehensive characterization of cell type anatomy in the mammalian brain to date, establishing a structural basis for building an Forebrain Projection Cell Atlas that integrates in-depth multi-modality anatomic and molecular information. The overall strategy is to establish a comprehensive mesoscale connectome among forebrain structures, then systematically label projection neuron subpopulations and cell types that underlie these pathways and finally obtain high-resolution whole brain datasets.

Anatomy Segment (in reference to Specific Aim 2 in Huang U19 grant proposal):

Systematically identify projection neuron subpopulations and cell types that underlie major forebrain pathways. We will combine genetic and viral labeling with Serial Two-Photon Tomography (STPT) to:

- quantify the distribution patterns of a comprehensive set of genetically labeled subpopulations
- characterize and quantify the axon projection pattern of these projection neuron subpopulations in several functional systems (e.g. sensorimotor system).

This work will generate a quantitative brain atlas of the distributions and axon projection patterns of molecularly defined cell classes and cell types within the Common Coordinated Framework (CCF) of the adult brain.

Introduction

This document describes the materials and methods used to process data generated from a Serial 2-photon microscope to quantify the molecular cell types

Metadata

For both cell-type CELL NUMBER and AXON PROJECTION atlasing, the brains are imaged using serial two-photon tomography (STPT) and the serial section datasets are acquired at x-y resolution = 1 micron and z resolution = 50 microns for 270-280 sections depending upon the rostro-caudal axial length of the brain.

The total size of uncompressed raw data is ~145 GB, after compression it is ~65 GB (LZW Lossless compression). Each high-resolution section is composed of 12 x 16 = 192 tiles, with each tile saved in 16-bit tif format.

Each dataset is associated with the following METADATA:

1) Cell type specific cre or flp mouse line (e.g Tle4, Fezf2 etc)
2) Cre- or flp- reporter for cell labeling (e.g Ai75, H2BGFP) or cre-to-flp conversion and viral tracer for axon projection mapping (e.g. LSL-flp + AAV-fDIO-EGFP for axon tracing)
3) Injection site coordinates (axon projection mapping)
4) Mice derivate strain (e.g. C57BL/6 etc.)
5) Sex (male or female)
6) Date of Birth
7) AGE (in weeks)
8) Date of perfusion
9) Date of imaging
10) Imaging machine id
11) Laser wavelength
12) PMT value for channel 1
13) PMT value for channel 2
14) Voltage under objective
15) Surface measured on piezo
16) Imaging plane measured on piezo
17) Total number of sections

18) Data processing server
19) Raw and processed data storage server location

STPT Image Reconstruction

Method

The reconstruction part is specific to Serial two-photon tomography (Ragan et al. 2012). The initial part involves image stitching of a mosaic set of image tiles from the Photomultiplier tube data. The PMT data is read in binary/TIFF format files and cropped along the edges where there are saturation artifacts from turning of the galvo mirrors. During this step, the average of all the image tiles is also calculated to determine the illumination field of the setup. [Supplementary Fig. 1]. The images can be stitched together by either specifying a predetermined overlap or using position information from the linear X and Y stages. With meta-information from the microscope, the image series data is converted grid image data and finally a sequence of stitched mosaic files. When mosaicking, the user has the option to blend the tiles at the overlap, or use a simple overlap. When stitching without blending, the stitching can also be done in the reverse order of imaging so as to avoid regions that have been photo bleached previously while imaging the previous tile. This also features generation of XML files for generation of input files for terastitcher and TrakEM2 tile configuration files (Fiji Plugin) based stitching. This process also creates a volume thumbnail for a quick inspection of the imaging quality and downstream process of image registration with reference atlas. A Fiji Module is used to view full resolution data sets.

Implementation

Setting up the raw data

- Check if all sections (.tiff) are saved in both ch1 (red channel) and ch2 (green channel).
- Remove ch3 (blank channel) tiffs acquisition data - by using the script - ***move_test.bat***
- Following the ch3 channel data deletion, transfer ch1 raw data and ch2 raw data to the IBM SONAS server.

Processing scripts setup

- Create a process folder with project ID in the processing server (e.g. brain2, brain3 or brain5)
    - /data/<user>/STP_processed_data/<cell_type>/<group1_male>/

- /data/<user>/STP_processed_data/<cell_type>/<group2_female>/
- Copy the reconstruction script in the process folder and input relevant data path and edit the runReconstruction_part1.sh script in gedit editor
- Update Mosaic text file is located in ch1 raw data folder (fig.1)
- Update the CH2_TIF_DIR and CH3_TIF_DIR (Ch2 tif directory path can be directly copied by right click copy command from the file browser nautilus)
- Update SECTION_STARTNUMBER and (fig.1) and SECTION_ENDNUMBER (e.g. 270) from the imaging.

```
##### BASIC USER EDIT START HERE ######################################################################

IS_TWO_CHANNEL=1 # put 1 for 2chnl and 0 for 1chnl raw data #
export CH1_PARAMFILENAME=/mnt/sonas/rpalanis/171011_RP_ChatAi75334939_raw/171011_RP_ChatAi75334939_raw_ch1/Mosaic_610f7332-a48a-4acd-b502-3494366c4ef0.txt
export CH2_TIF_DIR=/mnt/sonas/rpalanis/171011_RP_ChatAi75334939_raw/171011_RP_ChatAi75334939_raw_ch2/
export CH3_TIF_DIR=blank
export SECTION_STARTNUMBER=1
export SECTION_ENDNUMBER=270
```

Figure 1: Reconstruction script showing ch1 and ch2 file path entries along with section start and end numbers

Verify the advanced parameters: Values are set to default as shown in Figure 2.

- IMAGE_CORRECTION=1
- FUSION_METHOD=4 (no blending)

```
export OFFSETPOSITION=0
export NTHREADS=8
export IMAGE_CORRECTION=1  # put 1 for image correction and 0 for skip image correction #
export IMAGE_CORRECTION_METHOD=0  # put 0 for Matlab type and 1 for Fiji type #
export STITCHING=1  # put 1 for stitching and 0 for skip stitching #
export RESIZING_PERCENTAGE=0.05
export FUSION_METHOD=4  # put 0 for minimum, 1 for maximum, 2 for average, 3 for blending, 4 for no blending #
export CH1_PREFIX=ch1
export CH2_PREFIX=ch2
export CLIPWIDTHNORTH=15
export CLIPWIDTHSOUTH=15
export CLIPWIDTHEAST=15
export CLIPWIDTHWEST=15
export OVERLAP_PERCENTAGE=12
export STAGE_POSITIONS=0
```

Figure 2: Reconstruction script showing advanced stitching parameters including image correction and stitching blending method.

- Open Terminal. Change directory to the processed data folder. Run the script from Terminal command.

Level 0 Data description

The output of the reconstruction are the following files

- Stitched image directories of each of the imaged channels (Typically Red and Green) in stitchedImage_ch1 and stitchedImage_ch2. This contains the coronal section files at 1 µm resolution and 50 µm spacing in between sections. The stitched images are saved in lossless LZW compressed 16-bit tif images.
- The warping folder that contains the 5% downsized image stack that can be opened in ImageJ.

The intermediate data are the following:

- The average tile used for illumination correction for the each channel of the dataset is stored as averageTile_ch1.tif and averageTile_ch2.tif
- In Log/Mosaic_*.mat file contains all the meta information about the imaging parameters stored by the TissueCyte Scope:
    - Raw tile file names
    - Stage positions in each section
    - Imaging parameters like laser power etc.
    - Processing parameters, intermediate processing file names
    - Missing tiles during imaging

Image Reconstruction - Quality Control

- Visualize the stitched images in Fiji to check overall quality of the acquisition
- Obtained data folder structure should appear as shown below (fig. 3).

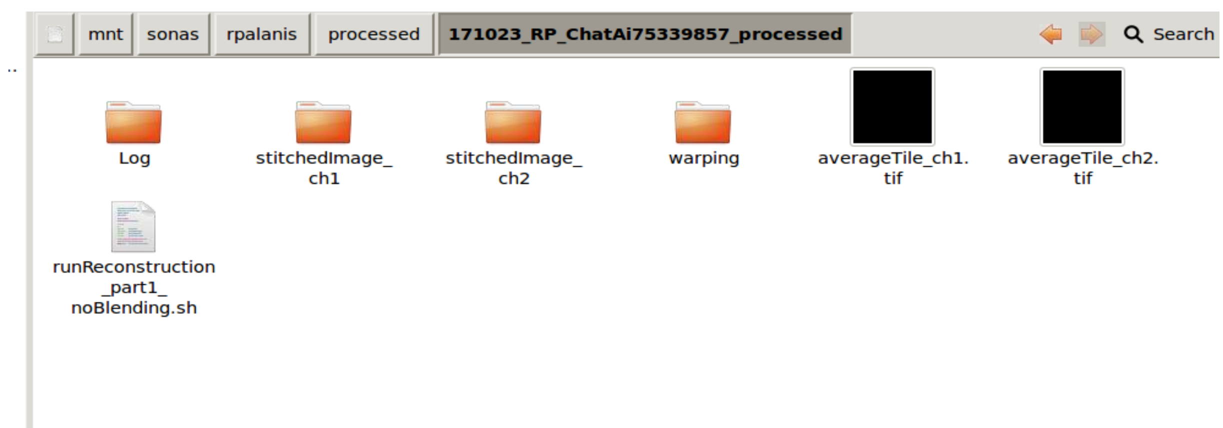

Figure 3: Processed data folder containing reconstruction script along with high-resolution channel 1 and 2 folders along with downsized p05 files in the warping folder.

-
    - Stitchedimage_ch1 and stitchedimage_ch2 folder contains high resolution acquired images.
    - The warping folder contains 5% downsized images and these images can be visualized for checking the overall quality of the imaging acquisitions.
    - Log folder has missing tiles info

- Few examples of the troubleshooting are below (Fig.4)

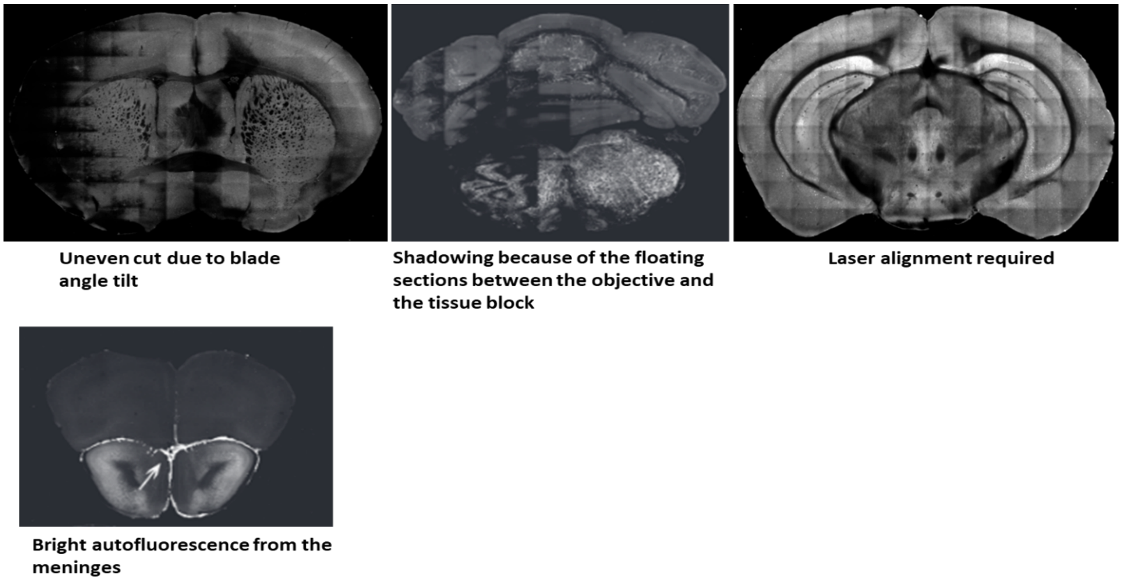

Figure 4: Examples of imaging artifacts detected after image reconstruction

Cell-type distribution and axon projection detection & Registration to CCF space

Method

Cell / axon detection

Convolutional neural networks (CNN) (Lecun et al. 1998) are used to detection of cells and other structures. Our segmentation tool has the capability to detect nuclear, cytoplasmic and axonal signals imaged using fluorescent light microscopes. The cell / axon detection CNN is built over CNS (Mutch et al. 2010) and CNPKG uses an older CUDA framework. These tools require an nvidia CUDA-enabled GPU to process these detections. The networks have both variable hidden layers and field of view depending on the feature that is being detected.

Registration to CCF space

This module registers the downsampled sample brain to CCF space. It is executed in 2 steps, first the affine and then followed by b-spline registration. The registration uses Mattes mutual information as similarity measure (Mattes et al. 2003). We use a toolbox for image registration call elastix (Klein et al. 2010). It is based on ITK (Ibáñez 2003; Niedworok et al. 2016)

Implementation

As described above there are multiple cell signals that can be identified. Cell detection network IDs and their applications are listed below:

- 164 – single channel quantification of c-fos positive cells and any nuclear based signals
- 10 – Mecp2-GFP or intra-nuclearly distributed signal based dataset quantification
- 313 – subtraction channel based quantification of c-fos and any nuclear based signals
- 320 – Rabies/Somatic labeling quantification

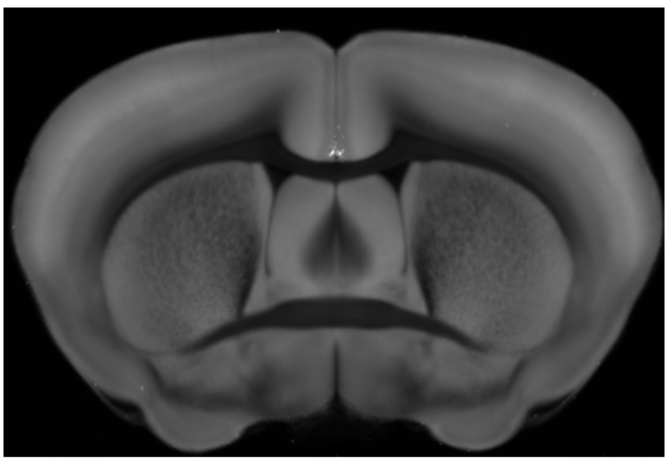

Figure 5: Shows coronal reference view of anterior commissure

The following steps are necessary to implement the signal detection networks:

- Identify anterior commissure region in the stitched images (fig.5)
- Calculate image intensity mean and standard deviation of the anterior commissure region using matlab for both the channels.
- Find the warpable range of the stitched brain by comparing the start and end coordinates with respect to the 25 micron Allen reference atlas.
- Input all the relevant values in the prediction script, change channel directories and run predictions.

Level 1 data description

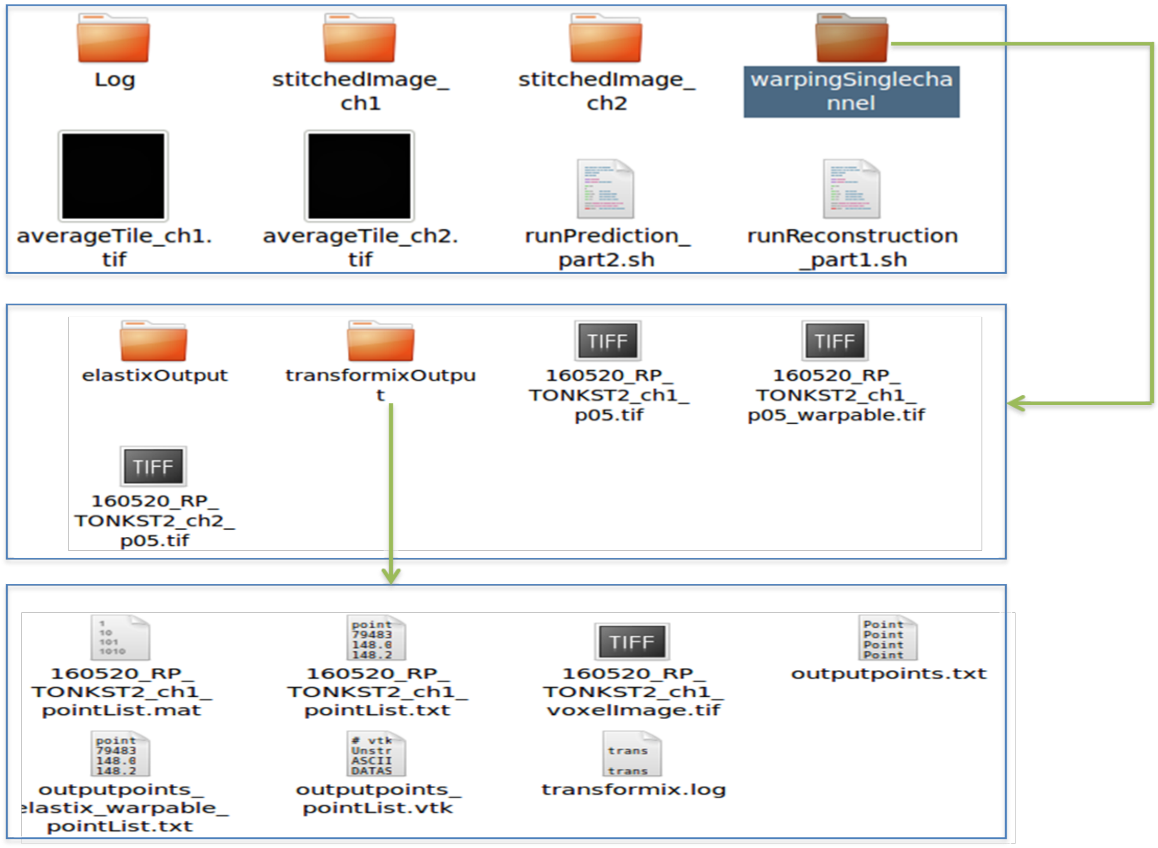

Figure 6: Processed folder showing the location of transformix Output folder contents generated after prediction.

- The detected cell masks can be found under the directory expressing the signal followed by network ID directory (e.g. stichedImage_ch1/164)
- The cleanup directory has the mask of the cells detected
- Cell counts for the predicted channel are contained in the transformix output as displayed in Figure 6 which can then be directly used as an input file for the ROI based statistics analysis.

Cell / axon detection & Registration Quality Control

- The cell mask images can be opened alongside the corresponding reconstructed sections and the quality of cell / axon detections can be qualitatively assessed using a synchronized mouse pointer
- For registration, open the result file of registration in the elastixOutput folder in warping with the fixed image (the downsized warpable image) and synchronize the stack slices and mouse pointers and look for congruous edges and corners in each of the image stacks.

CCF Region Wise Cell counts

Method

With the registration parameters the detected cells' coordinates are transformed to CCF space so that regionwide counts can be determined.

Implementation

A combination of C++ and R code

1. For ROI statistics create a **Project folder** in the processing server.
2. The Project folder should contain at least designated groups for brains and the matlab script.
3. Put all the point list text files in the appropriate groups, for example Group 1 can contain all the point list files from female brains and the same for male brains in Group 2.
4. Run the Matlab script – *runROI_all* in the processing server.
5. Extract all (right and left hemisphere) summary text file from the output folder and copy it in spreadsheet (*.csv) format to visualize region wise cell count.

Level 3 data description

The spreadsheet has the sample ID rows and the regions listed as columns. It also computed the mean and standard deviation for groups of animals

Data Compression for R24 Archival

After the reconstruction and cell / axon detection steps, the corresponding raw folders (the files containing the tiles from STPT computer) should be compressed using the script shown below in Figure 7. This script can be run on the processing servers without assigning the GPU since it is CPU-intensive in its processing.

Document the size of the raw data before compression, approximately 145 GB, and after compression, approximately 65-75GB.

There are provisions to convert the Data Level 1: reconstructed images (R and G channels) to JP2 file format. The L2 cell masks are compressed binary tif images.

```
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/local/MCR/v80/bin/glnxa64:.
export MCR_DIRECTORY=/usr/local/MCR/v80

export RAW_TIF_DIR=/mnt/sonas/rpalanis/171021_RP_ChatAi75334940_raw

cd $RAW_TIF_DIR
for folder in `ls -d -- */*/`; do
export fullDir=$RAW_TIF_DIR/$folder
export COMPRESS_EXE=/usr/local/prototype_matlab/V96/bin/run_compressTIFs.sh
$COMPRESS_EXE $MCR_DIRECTORY $fullDir 'tif'
echo $fullDir
done
```

Figure 7: Script showing the file path and directory information for microscope raw data compression

Dataset size

Data type	Data description	Usage Per Brain before compression	Usage Per Brain after compression	Usage per month with compression (20 brains)
Level 1 data	Stitched coronal section 1µmx1µm resolution spaced 50µm	~14GB	~25GB	~500GB
Level 2 data	Registration	<1GB	<1GB	~5GB
Level 3 data	Cell / axon detection masks	<1GB	<1GB	~5GB
Total Data	Raw & processed	~75GB	~25GB	~500GB

Level 0 (STPT raw data): Every month we expect to produce ~6 TB of raw, uncompressed STPT data (5 brains per week). After LZW compression, the data size will be reduced to 1.5 TB.

Levels 1,2,3: The typical size of a processed data folder which contains image reconstruction, cell/axon detection and registration is about 75 GB per brain, which amounts to 1.5 TB per month for 20 brains with OME-TIFF supported LZW compression. If JP2 is used then we can go down to 500GB